Walker Zachary Lee

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PROFESSIONAL SUMMARY

Computer Science graduate who builds full-stack web applications, AI-powered tools, and native iOS apps with JavaScript, Python, Swift, and the Claude (Anthropic) API. Comfortable across the stack — single-page front ends, Python/FastAPI backends with relational databases and JWT authentication, REST APIs, and OAuth integrations. Strong interest in workflow automation and applying generative AI to solve practical, real-world problems.

TECHNICAL SKILLS

Languages: JavaScript, Python, Swift, SQL, HTML, CSS
Frameworks & Tools: FastAPI, Flask, SQLAlchemy, SwiftUI, Xcode, Git & GitHub, Chart.js, Claude API, Gmail API, FFmpeg, Whisper, AWS
Concepts: REST APIs, JWT & OAuth 2.0 authentication, role-based access control, relational database design, native iOS development, prompt engineering, RAG, speech-to-text, AI workflow automation

CERTIFICATIONS

Google AI Essentials — Google, via Coursera (2026). View credential

PROJECTS

Full-Stack IT Support Desk — *Python · FastAPI · SQLAlchemy · SQLite · JWT · JavaScript*
Support ticketing platform built end to end: a FastAPI/SQLAlchemy REST backend over a SQLite database (Postgres-ready) with a vanilla-JavaScript single-page front end. JWT auth with bcrypt hashing, three roles (customer, agent, admin), full ticket lifecycle with threaded comments, live notifications, and an admin analytics dashboard driven by SQL aggregations. Hardened with role-based permissions, IDOR protection, Pydantic validation, and rate limiting.

BoxingTimer — Native iOS Boxing Timer — *Swift · SwiftUI · ActivityKit · Xcode*
Native iPhone and iPad round timer built from scratch in SwiftUI to my own design: brushed-metal styling, seven-segment LED displays, animated progress rings, and orientation-aware layouts. Adds a lock-screen Live Activity via ActivityKit, a custom colour-theme editor with persisted settings, and testing across multiple device sizes ahead of App Store submission.

AI Video Clipper — *Python · FastAPI · FFmpeg · Whisper · Claude API · ElevenLabs*
AI-powered video clipper that transcribes footage with Whisper, uses the Claude API to identify highlight-worthy segments, and renders finished clips with FFmpeg. Backed by a FastAPI service with GPU-accelerated processing and ElevenLabs voice generation.

Email Automation Bot — *Python · Gmail API · OAuth 2.0*
Auto-responder that authenticates to Gmail via OAuth and triggers contextual replies to incoming mail — demonstrating real API integration and secure authentication beyond the front end.

Portfolio FAQ Assistant — *JavaScript · DOM*
Rule-based, keyword-driven chatbot built from scratch and embedded in the portfolio site to answer visitor questions about my projects and skills.

Monthly Expenses Tracker — *JavaScript · Chart.js · localStorage*
Self-contained budgeting web app with custom categories, BHD currency, budget tracking, and live charts; persists locally and runs as a single file.

Product Page — *JavaScript · HTML · CSS*
Polished e-commerce product page with real colorway photography, interactive feature hotspots, and a mock checkout flow.

WORK EXPERIENCE

IT Specialist Intern — Infracorp (GFH), Bahrain — May 2025 – Jul 2025

- Network Management: Configured, maintained, and troubleshot both hardware and software across the IT environment.
- Cloud & Collaboration: Managed OneDrive and SharePoint for data storage, sharing, and team collaboration.
- System Administration: Handled user access control, software updates, and security across the IT infrastructure.
- Technical Support: Resolved connectivity, software, and hardware issues to keep teams communicating smoothly.

Intern — Sama Card Bahrain — Supported web development and digital marketing initiatives. — Oct 2023 – Dec 2023

Support — Freia Marketing — Produced content and managed social media channels. — Jan 2022 – Jun 2022

EDUCATION

University of Technology Bahrain — BSc in Computer Science. Varsity Basketball. — Sep 2019 – Dec 2024

Al Noor International School — Cambridge A Levels and IGCSEs. — Sep 2010 – Jun 2019